National University of Computer & Emerging Sciences, Karachi
Assignment-3
FALL 2026

Course Code: CS2009	Course Name: Design & Analysis of Algorithms
Max Marks	110
Date of Submission	13th Nov 2026

Group assignment (Submission as a group of 3 students)

- A **scanned copy of the handwritten assignment in PDF forma**t is required to be submitted to GCR within the deadline. A hard copy could be submitted in the next class, after the deadline. Typed Assignments will not be acceptable.
- 20% penalty for 1 day late
- 40% penalty for 2 days late
- Submission not allowed afterwards

Understand how these problems can be solved using dynamic programming and solve each of the following problems using an example.

a) The Coin Change Problem is a classic algorithmic problem that involves finding the number of ways to make change for a given amount of money (n) using a set of distinct coin denominations. Each coin denomination can be used unlimited times, and the goal is to determine how many combinations of coins can be used to make up the total amount. **Now solve for S = {1, 3, 5, 7}; the desired change is 12. [10 Marks]**

b) The Edit Distance, also known as the Levenshtein Distance, measures the similarity between two strings by calculating the minimum number of operations required to transform one series into another. The allowed operations are:

i. Insertion: Add a character to the string.
ii. Deletion: Remove a character from the line.
iii. Substitution: Replace one character with another.

The goal is to find the minimum number of operations needed to equal two strings. **Now solve for str1 = "SATURDAY" and str2 = "SUNDAY" .**

[10 Marks]

c) The Rod Cutting Problem is a classic optimization problem in computer science and mathematics. It involves finding the best way to cut a long rod into smaller pieces to maximize the total value of the details. Each piece of the rod has a length and an associated weight, and the goal is to determine the combination of cuts that yields the maximum value.

Now solve for length [] = {1, 2, 3, 4, 5, 6, 7, 8}, price [] = {1, 5, 8, 9, 10, 16, 18, 20}, Rod Length: 8. [10 Marks]

d) Given an input string s and a dictionary of words wordDict, determine if s can be segmented into a space-separated sequence of one or more words from wordDict. Words in wordDict can be reused. **Now solve for wordDict = {go, goal, goals, special, mobile, man, mango}. Input: "goalsmobile". [10 Marks]**

e) For Coin-change-making-problem, S = {1,3,4,7}, Desired Change is 10 j)
for wordDict = {go, goal, goals, special, mobile, man, mango}. Input: "goalsmobile."
[10 Mark's]

f) Take any real-world daily life task and write an algorithm to solve that using a dynamic programming approach that is mapped on one of the problems above. For example, You want to plan meals for a week while maximizing nutritional value within a budget. Each meal has a cost and a nutritional value. The goal is to select meals that fit within your budget and maximize total nutritional value. Also dry run the algorithm. [**20 Marks**]

g) For Longest-Common-Subsequence, X: {B, First Alphabet in the Name of Group Member 1, B, C, A, B} and Y: {First vowel in the Name of Group Member 2, D, A, B, A, B}

b) For Shortest-Common-Supersequence, X: First Name of Group Member 1 and Y: First Name of Group Member 2

c) For Longest-Increasing-Subsequence, {4th Digit of Group Member 1, 10, 2, 4th Digit of Group Member 2, 20} [**10 Marks**]

h) Matrix Chain Multiplication, p0 = 4, p1 = 6, p2 = 5, p3 = 20, p4 = 1, p5 = length of first name of Group member 3. Show parenthesis at the end. **[10 Marks]**

i) For the 0-1 knapsack problem, Value = [1, 4, 5, 7, 4], Weight = [1, 3, 4, 5, 2],
W = Number of characters in the first name of Group Member 1. [**10 Marks**]

j) For the partition problem, The Set contains 6 numbers. The first three numbers are the conversion of the first three alphabets of Group Member 1 into respective numbers, and the last three numbers are from the first three alphabets of Group Member 3.
For example: A = 1, B = 2, C = 3, etc. [**10 Marks**]